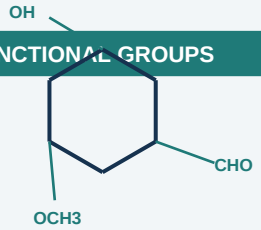
IDENTITY

C8H8O3 • 152.15 g/mol

IUPAC name: 4-hydroxy-3-methoxybenzaldehyde

STRUCTURE & FUNCTIONAL GROUPS



Schematic; substituent positions are indicated for concept.

- A benzene ring with three substituents
- Aldehyde group (–CHO)
- Phenolic –OH and methoxy –OCH3 groups

KEY GROUPS

Aromatic ring; aldehyde; phenol; ether

IMPORTANT PROPERTIES

- Usually a white to pale-yellow crystalline solid.
- Has limited water solubility compared with many organic solvents.
- The aromatic ring and oxygen-containing groups influence its odor, polarity, and intermolecular interactions.

WHAT IS AMAZING ABOUT IT?

One structure, several chemical roles

Vanillin combines an aromatic ring with an aldehyde, phenolic hydroxyl, and methoxy group. These groups affect both its characteristic aroma and how it dissolves or reacts in formulations.

USES & VALUE

- Flavor and fragrance ingredient
- Food, fragrance, and personal-care formulations
- Chemical feedstock and research compound

RISKS & SAFETY

- Avoid inhaling dust and avoid unnecessary skin or eye contact with concentrated material.
- Follow the supplier's safety data sheet and applicable food-grade handling rules.

CHEMICAL ENGINEERING CONNECTION

Vanillin manufacture and recovery connect reaction chemistry with extraction, crystallization, drying, and purity testing. Solubility and temperature affect separation choices.

SELECTED REFERENCES

National Center for Biotechnology Information. (n.d.). PubChem compound summary for vanillin (CID 1183). https://pubchem.ncbi.nlm.nih.gov/compound/1183
National Institute of Standards and Technology. (n.d.). Vanillin. NIST Chemistry WebBook. https://webbook.nist.gov/